

• 1. **Introduction** to the course and its objectives.

• 2. **Mathematical Preliminaries** including sets, functions, and limits.

• 3. **Calculus I** covering differentiation and integration techniques.

• 4. **Calculus II** focusing on advanced integration and applications.

• 5. **Linear Algebra** introducing vectors, matrices, and linear transformations.

• 6. **Differential Equations** exploring ordinary and partial differential equations.

• 7. **Probability and Statistics** covering basic concepts and distributions.

• 8. **Complex Analysis** introducing functions of complex variables.

• 9. **Topology** exploring the properties of spaces and continuity.

• 10. **Final Review** and preparation for the final examination.

• 11. **Advanced Topics** including a brief overview of modern mathematical research.

• 12. **Guest Lectures** by experts in various fields of mathematics.

• 13. **Research Projects** where students can explore topics in depth.

• 14. **Mathematical Software** training sessions for tools like MATLAB and Python.

• 15. **Mathematical Olympiad** preparation and participation.

• 16. **Mathematical History** and the contributions of great mathematicians.

• 17. **Mathematical Philosophy** and the foundations of the discipline.

• 18. **Mathematical Modeling** and its applications in science and engineering.

• 19. **Mathematical Computing** and the use of computers in solving problems.

• 20. **Mathematical Communication** skills and writing mathematical proofs.

• 21. **Mathematical Literature** review and discussion of recent papers.

• 22. **Mathematical Career Development** and advice on further studies.

• 23. **Mathematical Society** events and networking opportunities.

• 24. **Mathematical Travel** and visits to research centers.

• 25. **Mathematical Awards** and recognition of student achievements.

• 26. **Mathematical Conferences** and presentations by students.

• 27. **Mathematical Workshops** and intensive problem-solving sessions.

• 28. **Mathematical Seminars** and regular meetings for discussion.

• 29. **Mathematical Clubs** and extracurricular activities.

• 30. **Mathematical Outreach** and public engagement activities.